



भारतीय सूचना प्रौद्योगिकी संस्थान राँची
Indian Institute of Information Technology Ranchi
(An Institute of National Importance under an Act of Parliament)
Ranchi, Jharkhand, India

Assignment-I

Subject Name: Complex Analysis, Real Analysis and Linear Algebra (Mathematics-III)

Subject Code: MA2001

Instructions: Solve all the questions systematically and submit it by **5:30PM** on **29th September, 2025** to the respective faculty.

1. Find the bilinear transformation which maps $i, -i, 1$ to $0, 1, \infty$ respectively. Find the fixed and critical points of the resultant transformation.
2. Show that the transformation $w = \frac{i(1-z)}{(1+z)}$ maps the circle $|z|=1$ into the real axis of w -plane and the interior of the circle $|z|<1$ into the upper half of the w -plane.
3. Find the image of the family of the circles $x^2 + y^2 = ax$ where a is real, under the transformation $w = \frac{1}{z}$.
4. Show that $f(z)=u+iv$ is constant if $\text{Im}(f)$ or $\text{real}(f)$ is constant.
5. If $f(z)$ is harmonic function of z , then show that
$$\left\{ \frac{\partial}{\partial x} |f(z)| \right\}^2 + \left\{ \frac{\partial}{\partial y} |f(z)| \right\}^2 = |f'(z)|^2.$$
6. Show that the function $v(x, y) = e^x \sin y$ is harmonic. Find its harmonic conjugate $u(x, y)$ and hence, corresponding analytic function $f(z) = u + iv$ in terms of z .
7. Find the poles or singularity of the following functions. Also, classify them as isolated and non-isolated.
 - (i) $\cot \frac{1}{z}$ (ii) $\frac{1}{\cos z - \sin z}$ (iii) $\frac{e^{1/z}}{z^2}$ (iv) $\frac{(z-2)}{z^2} \sin\left(\frac{1}{z-1}\right)$ (v) $\frac{z^2+1}{\sin\left(2\pi/z\right)}$
8. Prove or disprove that e^{-1/z^2} has no singularities.
9. Find the residue of (i) $\frac{1+e^z}{\sin z + \cos z}$ at $z=0$. (ii) $\tan z$ at its poles (iii) $(1+z+z^5) \sin \frac{1}{z^2}$ at $z=0$.
10. Using Cauchy's residue theorem, evaluate $\frac{1}{2\pi i} \int_c \frac{e^{zt}}{z^2(z^2+2z+2)} dz$, where c is the circle $|z|=3$.
11. Evaluate $\int_c \frac{4-3z}{z(z-1)(z-2)} dz$, where c is the circle $|z| = \frac{3}{2}$, using Cauchy integral formula.
12. Use Cauchy integral formula to evaluate $\int_c \frac{\sin \pi z^2 + \cos \pi z^2}{(z-1)(z-2)} dz$, where $c: |z| = 3$.



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13. If $f(z) = u + iv$, is any analytic function of the complex variable z and $u - v = e^x(\cos y - \sin y)$, find $f(z)$ in terms of z .
14. Show that the function defined by $f(z) = \sqrt{|xy|}$, satisfies Cauchy-Riemann equations at the origin but is not analytic at that point.
15. Evaluate $\int_0^{1+i} (x^2 - iy)dz$, along the paths $y = x$ and $y = x^2$ separately. Integral value depends on path?
16. Prove or disprove that u and v are harmonic conjugates for any complex valued function $f(z) = u + iv$.
17. Evaluate $\oint_c e^{\cos \theta} \cos(\sin \theta - n\theta) d\theta$, where n is a positive integer and c is a unit circle.

*****DO SMILE*****